

Gradient Descent for Minimizing a Two-Variable Function

Introduction

In this homework, the Gradient Descent algorithm was implemented to minimize a convex two-variable function

$$f(x, y) = (x - 3)^2 + (y + 2)^2$$

which has a known global minimum at $(3, -2)$.

The objective was to study how different learning rates and initial points affect the convergence behavior of the algorithm, using both contour plots and 3D surface visualizations.

Effect of Learning Rate on Convergence

The learning rate (α) plays a crucial role in determining the speed and stability of convergence.

- **Small learning rate ($\alpha=0.05$)**

The algorithm converges slowly toward the minimum. The contour plots show very small steps, requiring many iterations to reach the optimal point. Although stable, this learning rate is inefficient.

- **Moderate learning rate ($\alpha=0.1$)**

This value provides a good balance between speed and stability. The algorithm converges smoothly and efficiently, reaching the minimum in significantly fewer iterations compared to $\alpha=0.05$.

- **Large learning rate ($\alpha=0.5$)**

In general, a large learning rate may cause overshooting or oscillatory behavior. However, due to the simplicity and convexity of the quadratic function used in this homework, the algorithm still converges very quickly. The contour plots show larger jumps toward the minimum.

Effect of Initial Point Choice

Two different initial points, $(0,0)$ and $(5,-5)$, were tested.

The results show that while the **initial point affects the trajectory and direction of the optimization path**, it does **not affect the final solution**. In both cases, the algorithm converges to the same global minimum $(3,-2)$, which is expected for a convex function with a unique minimum.

Observations on Stability and Efficiency

- Gradient Descent is highly stable for convex quadratic functions.
- Smaller learning rates ensure stability but reduce efficiency.
- Moderate learning rates provide faster convergence without instability.
- The combination of contour plots and 3D surface plots provides clear insight into how the algorithm moves step by step toward the minimum.

Conclusion

This experiment demonstrates that Gradient Descent is an effective optimization method for convex functions. The learning rate has a significant impact on convergence speed, while the initial point mainly influences the optimization path rather than the final result. Proper visualization helps in understanding both convergence behavior and algorithm stability.